Introduction to Programming

Lecture 19

VU
Virtual University

Random Access Files

Files

- open (file_name , mode) ;
- close ( ) ;

```
ifstream myFilePtr ;
myFilePtr.open ( “myFile” , ios :: in ) ;
```

Output File Stream

- ios :: app
- ios :: trunc
- ios :: ate

Read/write a character

get ( ) Read a character

put ( ) Write a character

,size

Number of characters to read

Delimiter

File Positions

File Position Pointer

tellg ( ) Function

myFile.tellg ( ) ;

Returns a whole number which tell you the position of the next character to be read from the file

tellp ( ) Function

myFile.tellp ( ) ;

Returns a whole number which tell you the position of the next character to be written in a file

For Positioning in the file

seekg ( ) ;

seekp ( ) ;

seekg ( )

Number of characters to move to

Starting point

```
filePtr.seekg ( long Num , ios :: origin ) ;
```

seekg ( )

```
seekg ( 10L , ios :: beg ) ;
seekg (10L , ios :: cur ) ;
seekg ( 10L , ios :: end ) ;
```

Example 1

```
#include<fstream.h>
main ( )
{
  int length ;
  ifstream inFile ( “myFile.txt” ) ;
  inFile.seekg ( 0L , ios :: end ) ;
  length = inFile.tellg ( ) ;
}
```

File

Name	city	Date-of-Birth
⋮	⋮	⋮
Jamil Ahmed Sukkur		10-10-1982
⋮	⋮	⋮

Rawalpindi

Merge Method

Original file

This is a text
data And needs
To be replaced

Empty file

NOT

seekg ( )

```
seekg ( 2201L , ios :: beg ) ;
```

fstream

```
fstream myFile ( “Sample.txt” , ios :: in | ios :: out ) ;
```

OR Function

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1

Example 2

This is an Apple

This is a Sample

get ( ) and put ( ) character in a file

```
myInputFile.get ( c ) ;
myOutputFile.put ( c ) ;
```

read ( ) and write ( ) Functions

Area in memory

Number of bytes to be read

read (char *buff , int count) ;

Area in memory

Number of bytes to be written

write (char *buff , int count) ;

Example 3

```
char str [ 10000 ] ;
myInputFile.read ( str , 10000 ) ;
myOuputFile.write ( str , 10000 ) ;
```

seekg ( )

```
seekg ( 0L , ios :: end ) ;
```

seekg ( )

```
seekg ( -1L , ios :: end ) ;
```

seekg ( )

```
seekg ( -2L , ios :: cur ) ;
```

Address of the integer 'i'

Number of bytes to be written

```
myOutputFile.write ( &i , 4 ) ;
```

sizeof ();

Address of the integer 'i'

Size of integer

```
myOutputFile.write ( &i , sizeof ( i ) ) ;
```

```
for ( i = 0 ; i < 100 ; i ++ )
{
 myOutputFile.write ( &i , sizeof ( i ) ) ;
}
myOutputFile.close ( ) ;
```